

Harshal Singh Vats

Ranchi, Jharkhand, India | harshalsvats@gmail.com | 732-003-3794 | linkedin.com/in/harshal-singh-vats | github.com/HarshalSinghVats

EXPERIENCE

Artificial Intelligence Research Intern

Jun 2025 – Aug 2025, Remote

JustTutors

- Built an AI-powered virtual tutoring system integrating speech-to-text, text-to-speech, and content delivery APIs.
- Built the backend using FastAPI and Python, integrating Groq and Gemini APIs for LLM inference and pyttsx3 for text-to-speech output to automate tutoring session handling and query routing.
- Contributed to model evaluation and performance benchmarking during internal pilot testing.

Full Stack Developer Intern

Mar 2025 – Jun 2025, Remote

EdEarn

- Built and integrated REST API endpoints in Node.js with responsive components designed in React, handled debugging and optimized backend request handling.
- Designed a MongoDB schema to store user session data and query logs, enabling fast retrieval of data across requests.
- Shipped features as part of a remote team using Git-based version control, with regular code review cycles.

SKILLS

Languages: Python, C, C++, JavaScript, Golang, SQL

Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS

Databases: MySQL, PostgreSQL, MongoDB

Frameworks: React, Node.js, Flask, Express, FastAPI

Libraries: pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

PROJECTS

mlweb

Sarala Birla University • ml-web-main.vercel.app

- Developed a no-code web platform for building and training machine learning models through an interactive interface.
- Reduced manual ML setup time by ~50% by automating preprocessing and dataset handling pipelines.
- Implemented data visualization and exploratory analysis tools using Matplotlib and Seaborn.

Weblante

RanchiHacks • github.com/HarshalSinghVats/weblante

- Built a real-time browser extension for website safety analysis and content classification.
- Implemented hybrid rule-based and ML-driven classification, achieving 70% accuracy on unknown websites.
- Integrated fuzzy-logic based heuristic validation to reduce unsafe override behavior.

EDUCATION

Bachelor of Computer Applications

2023 – 2026

Sarala Birla University, Ranchi, India

- **CGPA: 8.77**
- Coursework: Data Structures, Algorithms, Object Oriented Programming, Computer Networks, Database Management Systems, Data Science.

INVOLVEMENT

Team Lead — Smart India Hackathon

Sep 2025

Ranchi, Jharkhand

- Led a team of 5 engineers to build a healthcare accessibility platform for the district of Nabha, enabling virtual doctor consultations, self-diagnosis support, and medicine delivery for remote villages.
- Designed community health worker support stations to assist residents of low-connectivity rural areas with healthcare access while creating local employment through support and delivery operations.

Head of Machine Learning Club

Oct 2024 – Jun 2025

Google Developer Group on Campus, Sarala Birla University

- Led ML initiatives and workshops for 100+ students across multiple sessions by inviting speakers to teach about dedicated topics and guide fellow peers.
- Mentored 20+ students entering the machine learning domain through structured learning paths and 1:1 communications.